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Finding Data

(and Academic Literature)

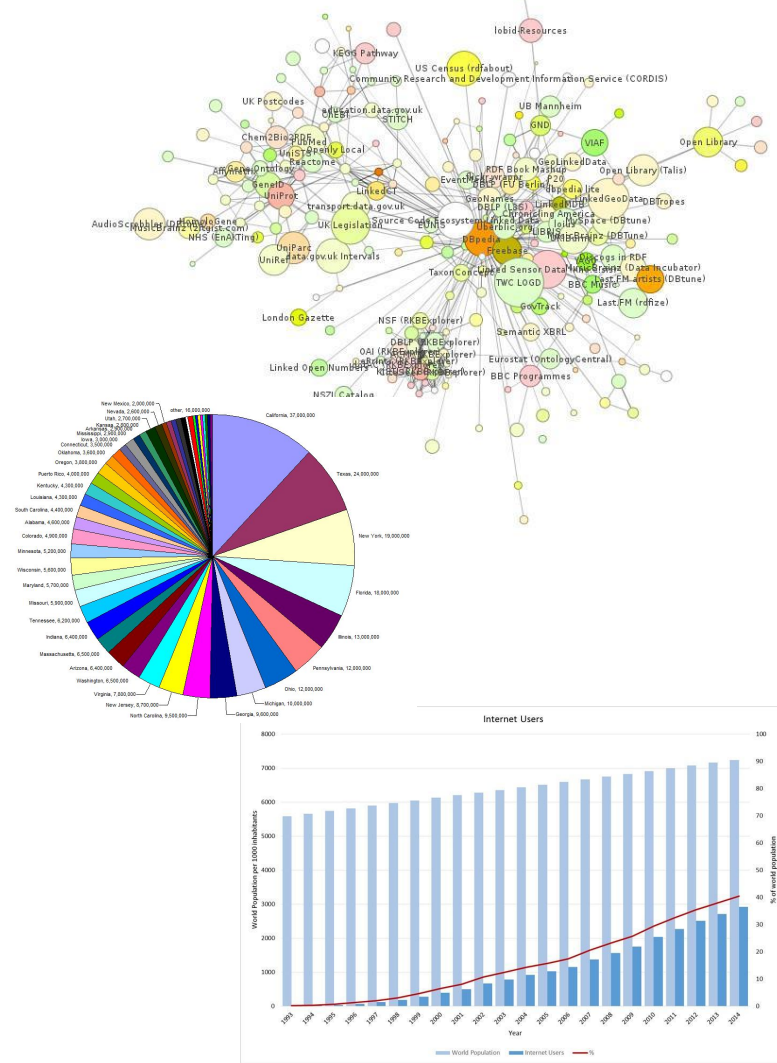
Session Guide:

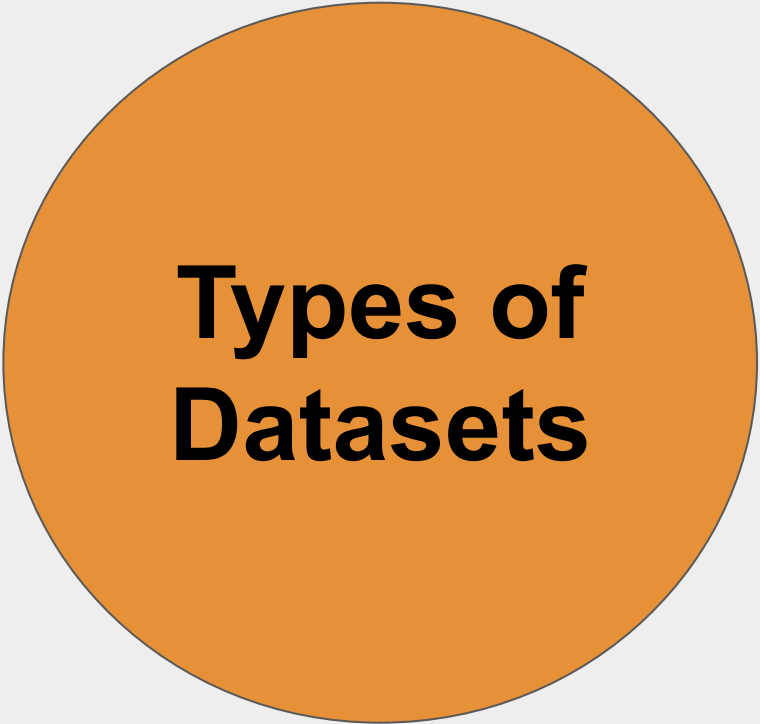
<https://libguides.colorado.edu/findingdata/2026>

Session Topics:

- Types of datasets
- Finding datasets
- Evaluating datasets
- Finding scholarly articles
- Licensing issues with datasets

Where did you find them?





Types of Datasets

- **Structured**
 - fixed schema with tables (rows and columns)
 - relational databases
- **Semi-structured**
 - No formal schema
 - JSON, XML, CSV files
- **Unstructured**
 - No formal structure
 - Text, images, videos, social media, sensor data



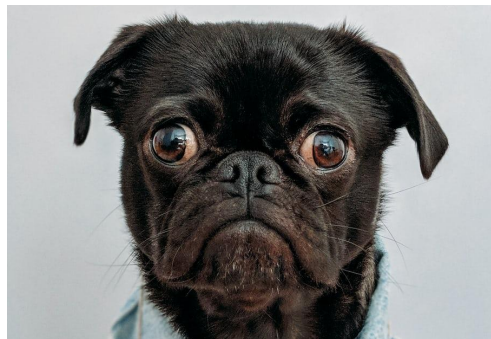
Big Data

- **Datasets that are:**
 - **Very large**
 - **Created or recorded quickly**
 - **Vary in structure**
 - **Cloud-based storage**
- **Contact [Research Computing](#) for large scale datasets and tools**



Finding Datasets

Manage Expectations



The perfect dataset...

1. may not exist
2. may not be free
3. may not be “good” or useable
4. may need significant cleaning
5. you may not be allowed to use it

The Library of Missing Datasets by Mimi Onuoha

"Missing data sets" are the blank spots that exist in spaces that are otherwise data-saturated...something does not exist, but it should...Spots that we've left blank reveal our hidden social biases and indifferences.

—Mimi Onuoha



(File being grabbed is titled: “Publicly available gun trace data”)

Define your project

1. What is your research question?
2. What (exactly) are you studying?
3. Which discipline(s) are relevant?
4. Time period & location?
5. How much data do you need?
6. How much data can you handle?
7. What software is available to you?
8. Where will you store your data?
9. Has anyone else done a similar project?



Who Collects the Data?

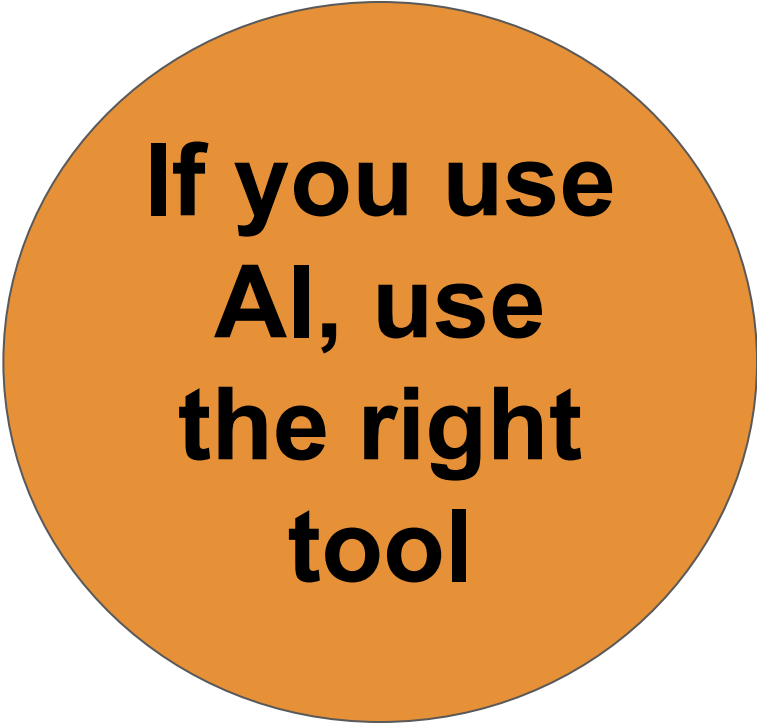


(takes time, money, effort)

- a. Specific researchers / labs?
- b. Government departments and agencies? (EPA, Census Bureau?)
- c. International organizations? (World Bank, WHO?)
- d. Companies? (Facebook, Amazon, Pfizer, etc.?)
- e. Universities?

**Have you used AI to find
datasets?**

Any issues?

An orange circle with a thin black outline, containing the text "If you use AI, use the right tool" in bold black font.

**If you use
AI, use
the right
tool**

- LLMs tend to hallucinate
- Not good at data and numbers
- Some tools like OpenAI's [Dataset Finder](#)
- Consider using [Kaggle](#), [Perplexity](#), [Consensus](#), or other academically focused AI tools

A large orange circle with a thin black outline, containing the text 'Other ways to find datasets' in bold black font.

Other ways to find datasets

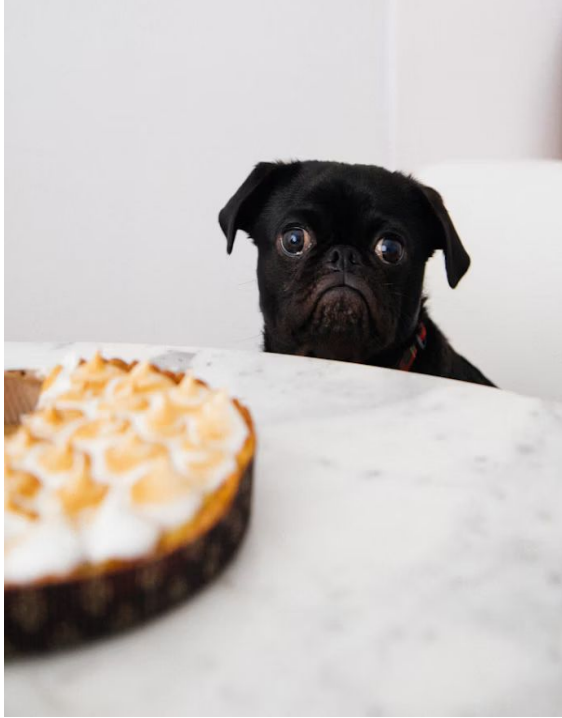
- Data repositories
- Academic articles
- Government websites
- International data hubs
- Ask a librarian

Examples of Data Repositories

- [Kaggle](#)
- [Data.gov](#)
- [Earthdata.nasa.gov](#)
- [Microsoft Research Open Data](#)
- [Reddit Datasets](#)
- [ICPSR \(Inter-university Consortium for Political and Social Research\)](#)
- [World Bank Open Data](#)
- [World Health Organization Data](#)
- [Dryad](#)
- [Amazon Web Services \(AWS\) Data Exchange](#)
- [Data.europa.eu](#)
- [Figshare](#)
- [Zenodo](#)
- [CU Scholar](#)

Repository Registries (from [Finding Data Guide](#))

- [Google Dataset Search](#) *Dataset Search is a search engine for datasets. Using a simple keyword search, users can discover datasets hosted in thousands of repositories across the Web.*
- [re3data.org](#) *Re3data is a global registry of research data repositories that covers research data repositories from different academic disciplines.*
- [Open Access Directory's List of Open Repositories](#) *This is a list of repositories and databases for open data.*
- [Open Access Directory's List of Repositories by Discipline](#)
- [FAIRsharing.org](#) *A registry of knowledgebases and repositories of data and other digital assets.*
- [NIH-Supported Data Sharing Resources](#) *To help researchers locate an appropriate repository for sharing or accessing data, BMIC maintains lists of domain-specific and generalist data sharing repositories that are each described by several properties (See Repository Properties Definitions below).*
- [PLOS One's list of recommended data repositories](#) *PLOS (Public Library of Science) has identified a set of established repositories, listed below, that are recognized and trusted within their respective communities.*
- [Paperswithcode: Machine Learning Datasets](#)



You found the perfect dataset.

But, can you access it?

Open Data (publicly available)

- [2022 OSTP Memo](#):
- Office of Science and Technology Policy (OSTP)
 - Office of Science and Technology Policy (OSTP) requires researchers to provide free, immediate (without embargo), and equitable access to research that is federally funded
 - Applies to:
 - all research paid for by federal agencies
 - Datasets and papers must be open

Proprietary Data (no public access)

- Protected by copyright, patents, contracts
 - Software/coding
 - Business or financial data
 - Recipes, designs, music, art
- Protected for medical or other sensitive information
 - insurance data, health data, data protected by court order

(Sometimes access can be purchased)

Evaluating Datasets



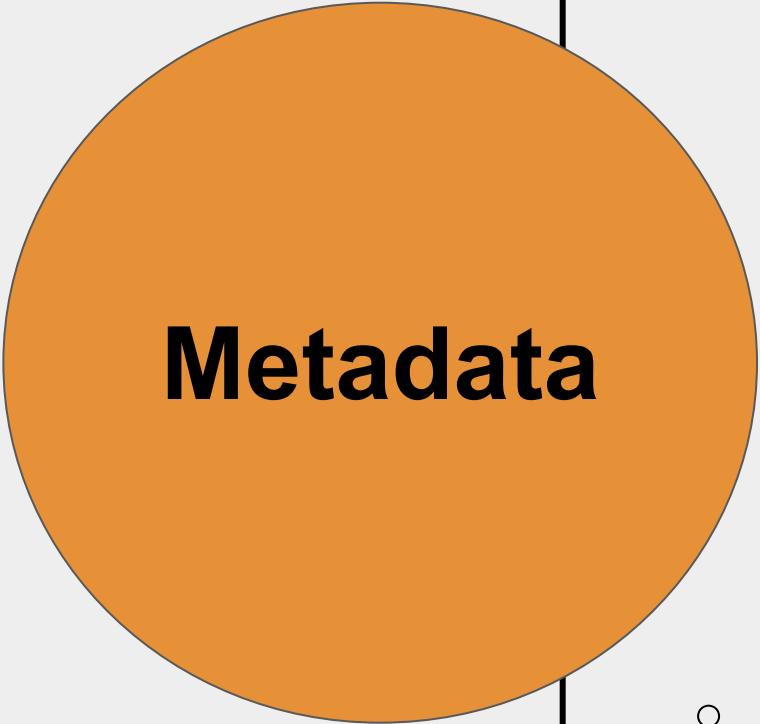
What is “good” data anyway?





Ethical Issues

- Data is often about people, places, animals
 - Their safety, privacy, and welfare are important
- Who owns the data?
 - Do you have permission to use it?
- Is there personal information in the dataset?
- Who could be harmed by the data?
- Consider motives of whoever collected, funded, compiled, and published the dataset
- For more information:
 - [Handbook of the Modern Development Specialist](#)
 - [CU Boulder Human Research Office](#)



Metadata

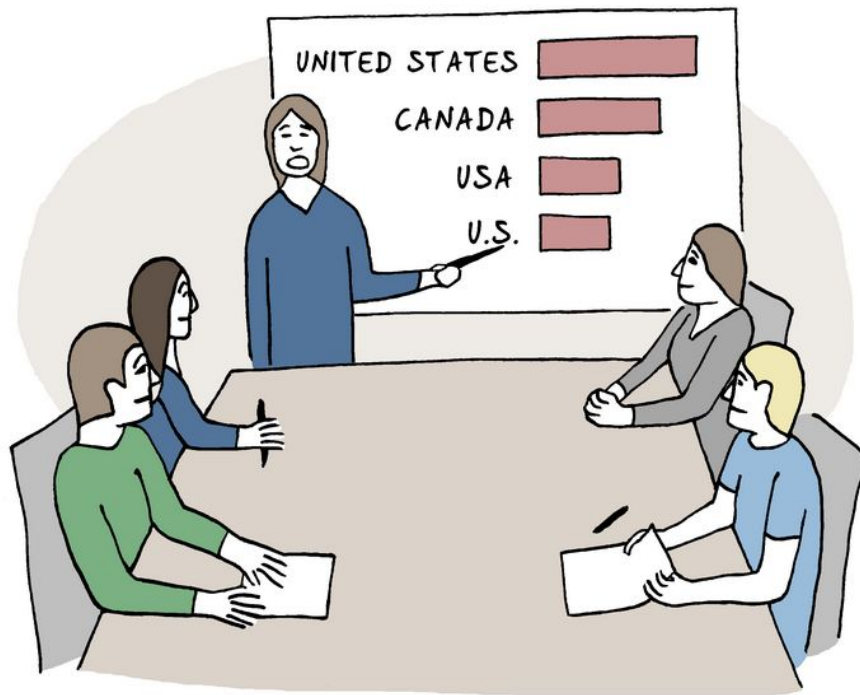
- **Includes:**
 - Licensing information
 - Technical requirements
 - The who, what, where, when, why and how the data was created
 - Units measured, labels, inconsistencies or issues to know about
- Where to find it:**
 - Readme file, data dictionary, codebook, attached file, repository page
- **Why is it important?**
 - Helps you use and understand a dataset

Metadata Standards

- Guidelines to help dataset creators provide consistent information about their dataset
- Discipline-specific requirements
- Help others understand and use a dataset

A Few Examples of Metadata Standards

- [Astronomy Visualization Metadata](#)
- [Darwin Core](#)
- [Data Documentation Initiative \(DDI\)](#) to document numeric data files
- [Dublin Core](#), a general purpose metadata standard
- ISO 19115 or FGDC's [Content Standard for Digital Geospatial Metadata](#) for geospatial data
- [Ecological Metadata Language](#)



AS YOU CAN SEE, OUR TOP MARKETS ARE
UNITED STATES, CANADA, USA AND THE U.S.

Datasets in tutorials



Datasets in the wild



[Image credit NC233](#)

“Good” Datasets

1. **Complete**
2. **Require minimal cleaning**
3. **Explain data collection methods**
4. **Clear labels (i.e., variables, column headers)**
5. **Declares bias**
 - a. **Conflicts of interest**
 - b. **Source of funding**
6. **License Information**
7. **Ethical & Protects privacy**
8. **Usable Format**



“Bad” Datasets

- 1. Incomplete / errors**
- 2. Require cleaning**
- 3. Outdated**
- 4. No / poor documentation**
 - a. No info about context or methods**
 - b. Poor metadata**
- 5. Unethical / biased**
- 6. Too much data**
- 7. Incompatible with software**



Evaluating Dataset Checklist:

1. Is the dataset:

- a. Usable: readable, well-documented, and available (to you)
- b. Functional format for software/analysis
- c. Complete, has good metadata (readme file!)
- d. Minimal “cleaning” or “wrangling” needed
- e. Data is current

☐☐☐

2. Does it follow a metadata standard?

3. How was the data set created and why?

4. What kinds of bias or issues exist in the dataset?

5. Could the use of the dataset be harmful in some way?

6. How has the dataset been used? How could it be used?

Text and Data Mining (TDM)



Photo by NeedPix

Answers

- Contact libraries@colorado.edu for specific questions about the use of licensed content in Libraries' electronic collections.
- Look for relevant licensing terms in Libraries discovery systems (e.g. OneSearch).

If you can't find a dataset:

- Contact your librarian
- Contact the [Center for Research Data and Digital Scholarship](#)



Academic Literature: Why Bother?

[Image](#) by Jongsun Lee on Unsplash

How Scholarly Literature Can Help

1. Provides you with:
 - a. Context/history of your topic
 - b. Knowledge of **terms used by experts** in the field you are investigating
 - c. Can help you **identify places to search for data**
2. Shows you what research has already been done on your topic
 - a. See the methods used to collect and analyze data
 - b. See what issues investigated (so you don't repeat what has been done)
3. Sometimes you can find data sets in articles or in databases
 - a. Web of Science allows you to search for articles with data included
 - b. ACM (calls data "[artifacts](#)")

Where to Find Scholarly Articles

- Library databases: we have [Library Guides for every subject/discipline](#)
 - Some popular ones:
 - [Web of Science](#)
 - [Academic Search Ultimate](#)
 - [Association of Computing Machinery \(ACM\)](#)
 - [PubMed](#)
 - [Google Scholar](#) (connect through Libraries!)
- AI: but watch out, verify articles and actually look at them to avoid a “bad” article. Use scholarly AI more than the typical LLMs (Chat, Gemini, etc.)



Library databases are different from Google & AI. They use “search terms” (specific word that represent your topic) instead of natural language queries.

(Your librarian can help you use them well!)

Use Citation Management Software!

- [Zotero](#)
- EndNote
- Mendeley
- EasyBib
- RefWorks



For help on citations, contact your librarian or consult Purdue's [OWL \(Online Writing Lab\)](#)

Please feel free to contact me:
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